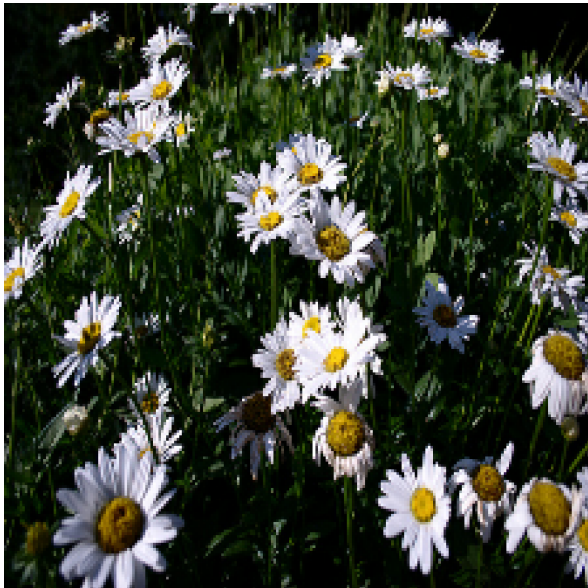
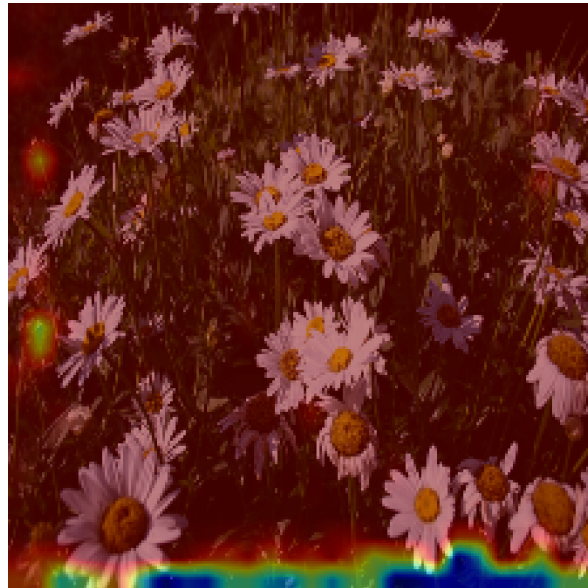


Input

 M_S^{P1}  M_S^{P2}  M_S^{P3}  M_S^{P4} 